

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0645

Krumkill 100 kV
May 20, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0645. The Interconnection Customer proposes a 151.7 MW solar photovoltaic facility interconnecting at the Krumkill 100 kV substation in the Illinois area. The request was received on July 15, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 332 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,130,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: DAVEY TOWN - KRUMKILL 100 kV #1
Contingency: KRUMKILL - DAVEY TOWN 100 kV Ckt 3
Loading: 106.3% (Rating: 151 MVA)
DFAX: 22.9%
Pre-project loading: 83.4%
2. Facility: KRUMKILL - DAVEY TOWN 100 kV Line 2
Contingency: KRUMKILL - DAVEY TOWN 100 kV
Loading: 106.3% (Rating: 151 MVA)
DFAX: 22.9%
Pre-project loading: 83.4%
3. Facility: DAVEY TOWN - KRUMKILL 100 kV Cir. 3
Contingency: KRUMKILL - DAVEY TOWN 100 kV #1
Loading: 106.3% (Rating: 151 MVA)
DFAX: 22.9%
Pre-project loading: 83.4%
4. Facility: DELL - GUAYAMA PDS 100 kV Circuit 1
Contingency: GUAYAMA PDS - DELL 100 kV Ckt 2
Loading: 102.1% (Rating: 165 MVA)
DFAX: 26.4%
Pre-project loading: 77.8%
5. Facility: DELL - GUAYAMA PDS 100 kV Line 2
Contingency: GUAYAMA PDS - DELL 100 kV #1
Loading: 102.1% (Rating: 165 MVA)
DFAX: 26.4%
Pre-project loading: 77.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DAVEY TOWN - KRUMKILL 100 kV Ckt 1	Reconductor line	\$934,000	67% (\$629,000)
DAVEY TOWN - KRUMKILL 100 kV Circuit 2	Reconductor line	\$536,000	67% (\$361,000)
KRUMKILL - DAVEY TOWN 100 kV	Reconductor line	\$536,000	67% (\$361,000)
DELL - GUAYAMA PDS 100 kV Circuit 1	Reconductor line	\$797,000	64% (\$512,000)
DELL - GUAYAMA PDS 100 kV Ckt 2	Reconductor line	\$415,000	64% (\$267,000)
	Total allocated to project		\$2,130,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.